

Button and Fold a Dress Shirt SOP

One episode closes one long sleeved dress shirt and folds it. The shirt is already lying face up and spread on the bare desk when recording starts. The five snaps down the front placket are pressed closed from the collar down, the placket is pulled straight under tension between both grippers, the shirt is lifted by both shoulders and draped over face down, each side is folded in and its sleeve laid back behind the body, the shirt is turned to lie horizontal with the collar to the right, the body is folded into thirds, and the shirt finishes as one squared packet with the whole collar showing face up at the right end.

Every press, pull, and fold happens directly on the bare desk, inside the **work zone** at the center. The bare surface lets cloth skate, so the shirt is held still by a **pin** or held taut between both grippers whenever a gripper moves along it.

The shirt closes with **press snaps**, not sewn buttons. A snap closes on one straight downward press and nothing else. No snap is pinched shut between the gripper fingers and no placket is pulled sideways to bring a socket onto its stud.

Once the snaps are closed they stay closed. Nothing rubs down the placket and nothing pulls across it. The placket is straightened by pulling its two ends apart **along** the collar to hem line, which loads the closure the way it is built to hold, and the shirt is turned by draping it over from the shoulders rather than by any motion that peels the two placket edges apart.

The order never changes: fasten, tension, turn, fold the sides, rotate, fold in thirds. No side is folded before all five snaps are closed and the placket has been pulled straight, nothing is folded while the shirt is still face up, and the thirds are not folded until the shirt lies horizontal.

The right gripper does the primary work: it lays and presses every snap and folds the right side. The left gripper supports: it pins the cloth for every press and folds the left side. Both grippers work together, one end each, whenever the shirt is tensioned, turned, rotated, or folded into thirds. Each arm keeps to its own side of the desk and neither takes the other's single handed jobs.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.

2. The environment camera shows the whole work zone with the spread shirt inside it: collar, hem, both sleeves, and both cuffs in frame, with clear desk all round.
3. The environment camera also shows the space above the work zone, so the shirt stays in frame while it hangs from both shoulders during the turn.
4. The overhead view shows the front placket end to end, with all five snaps readable.
5. Both arms are at home with grippers open.
6. The desk is clear and bare apart from the shirt.
7. The left gripper reaches the left half of the work zone, the left shoulder, the left cuff, the collar band, and the placket at the center without the arm leaning out or reaching a joint limit.
8. The right gripper reaches the right half of the work zone, the right shoulder, the right cuff, the hem at the center, and the placket at the center without the arm leaning out or reaching a joint limit.
9. Both grippers together can lift the shirt by its two shoulders until the hem hangs clear of the desk, and can carry that hold forward and down through the drape without either arm reaching a joint limit.
10. With the shirt lying horizontal after the rotation, the left gripper reaches the far hem corner and the right gripper reaches the near hem corner at the left end, without either arm leaning across the other.

Materials checklist

1. One **long sleeved dress shirt** already lying face up and square on the desk, collar toward the far side and hem toward the front edge, inside the work zone with clear desk all round.
2. Both sleeves lie flat on the desk, angled out from the shoulders toward the front corners, cuffs flat and clear of the placket. Neither sleeve lies across the front.
3. The **over placket** rests on the **under placket** with all five snaps open. The over placket comes from the right side of the shirt, the under placket from the left.
4. The shirt closes with five **press snaps** evenly spaced down the placket from the collar to the hem. There are no sewn buttons and no buttonholes. The under placket carries the studs facing up, the over placket carries the sockets facing down.
5. Every snap closes on one straight downward press and holds. A snap that will not click closed, that pops open on its own, or that lets go when the closed shirt is lifted by its two shoulders, is replaced before the episode.
6. The shirt has no collar snap and no cuff snaps. The five placket snaps are the whole of the fastening work.
7. The shirt is pressed flat, dry, and free of stains, tears, and open seams. It holds a fold rather than springing open.
8. The shirt still carries no fold line from a previous episode.

9. The shirt is light enough to hang from its two shoulders without either gripper slipping, and short enough that the hem clears the desk when it hangs.
10. The desk surface is dry, clean, and free of anything the cloth can catch on.

Workspace layout

- **Work zone:** the center of the bare desk, reachable by both arms. It holds one shirt. Every press, every pull, and every fold happens here, and it is the only place a gripper opens.
- **Left half of the work zone:** the left body edge, the left shoulder, the left sleeve, and the left cuff. The left gripper's zone.
- **Right half of the work zone:** the right body edge, the right shoulder, the right sleeve, and the right cuff. The right gripper's zone.
- **Center of the work zone:** the placket, the collar, and the hem at the middle. Both grippers reach it, one to pin and one to work, and neither carries anything across to the far side of it.
- **Turn space:** the clear air directly above the work zone, where the shirt hangs from both shoulders during the turn. Nothing else stands on the desk under it.

Vocabulary

- **Arm assignment:** the right gripper lays and presses every snap and folds the right side and right sleeve. The left gripper pins the cloth for every press and folds the left side and left sleeve. Both grippers work together, one end each, to tension the placket, turn the shirt, rotate it, and make the thirds folds. Neither gripper takes the other's single handed jobs.
- **Front:** the side of the shirt carrying the placket. **Face up** is the front toward the ceiling, **face down** is the front against the desk.
- **Placket:** the doubled strip running down the front from the collar to the hem. The **over placket** lies on top and carries the sockets, the **under placket** lies beneath and carries the studs.
- **Snap:** one two part press fastener on the placket, a **stud** on the under placket and a **socket** on the over placket. The snaps are numbered from the collar down: snap 1 is nearest the collar, snap 5 nearest the hem.
- **Closed snap:** the two placket edges hold together when the gripper lifts off, with the edges in line, no gap, and no step between them. Nothing else counts as closed.
- **Press:** the closed gripper set straight down flat on the cloth, held still for 2 seconds, and lifted straight off. A press never slides, never rubs, and never pinches.
- **Pin:** a closed gripper pressed straight down on the cloth to hold it still. A pin never pinches the cloth and never travels.
- **Tension:** both grippers pinch the two ends of the same line on the cloth, come to the same height, and draw apart in opposite directions **along** that line until the cloth between them hangs straight with no slack. A tension pull never goes across the line it is straightening.

- **Stroke:** the closed gripper pressed down on the cloth and drawn along in one continuous run, end to end, without lifting off. A stroke is used on a fold edge or a sleeve on the **back** of the shirt. Nothing is ever stroked down the placket.
- **Shoulder:** the corner of the shirt where the sleeve seam meets the collar band, one on the left and one on the right.
- **Drape:** the turn. Both grippers hold the two shoulders, lift until the shirt hangs clear of the desk, carry the hanging body toward the operator and down, then move back and down so the shirt lays itself over onto its front and lands flat. The turn comes from where the arms carry the hold, never from twisting the wrists.
- **Settle shake:** one short up and down move of both arms while the shirt hangs from its shoulders, to let the body and sleeves drop straight. One shake, not a whipping fling.
- **Fold:** a gripper pinches an edge or a cuff, lifts it clear of the cloth below, carries it over, lowers it onto its target, and releases. Nothing is folded by pushing it along the desk.
- **Fold edge:** the closed folded side a fold creates, as opposed to the loose open edges.
- **Body edge:** the side seam line of the shirt, running from the shoulder to the hem.
- **Sleeve laid behind:** the sleeve lying flat on the back of the shirt, inside the side fold edge, running toward the hem end with the cuff pointing at the hem.
- **Rotation:** both grippers take the collar end and the hem end of the folded body, lift it clear, swing it round a quarter turn on the desk together, and lay it down horizontal. The turn comes from the two arms swinging the shirt round, never from twisting the wrists.
- **Horizontal:** the shirt lying across the desk with the collar at the right end and the hem at the left end, its long axis running left to right in front of the operator.
- **Thirds fold:** two folds that bring the hem end across toward the collar in even thirds, the second one stopping just short of the collar.
- **Packet:** the shirt once both sides and both thirds folds are done. One squared rectangle of layers with the whole collar showing face up at the right end.
- **Settled fold:** the cloth stays put for 2 seconds after the gripper lifts clear, with no layer springing open.
- **Release point:** the work zone on the desk. A gripper opens nowhere else.

Steps

Steps 1 to 7 run once, in order, on the one shirt. Step 8 ends the episode.

Step 1: Press the five snaps closed

Goal: all five snaps closed from the collar down, with the two placket edges in line.

- Work the snaps in order, snap 1 first and snap 5 last. A snap is closed before the next one is started.

- The **left gripper**, closed, **pins** the **under placket** just left of that snap's **stud** and holds it there until the press is finished.
- The **right gripper** pinches the **over placket** at that snap, lifts it just clear of the cloth below, lowers it so the **socket** sits on the **stud**, and releases.
- The **right gripper**, closed, **presses** straight down flat on the snap, holds for 2 seconds, and lifts straight off.
- The left gripper lifts clear only after the right gripper is off the cloth.

Lift the over placket clear and lower it onto the stud. A placket pulled sideways into line drags the whole front with it and lands the socket off its stud. The whole garment skates on the bare desk, so the left gripper's pin is what keeps the under placket still.

Check: the snap holds together when the gripper lifts off, the two placket edges are in line at that snap with no gap and no step between them, and no snap already closed has popped open. If the snap did not take, lift the over placket clear, lower it onto the stud again, and press again. If it still will not close after three presses, the snap is defective: stop, log it as a system issue, and discard the episode.

Step 2: Pull the placket straight under tension

Goal: the placket running straight from the collar to the hem, flat and free of rucks, with all five snaps still closed.

- The **left gripper** pinches the front at the **collar band** on the placket line, at the far end of the shirt.
- The **right gripper** pinches the front at the **hem** on the placket line, at the near end of the shirt.
- Both grippers come to the same height, just clear of the desk, and draw apart in opposite directions **along** the collar to hem line until the placket between them hangs straight with no slack. Hold the tension for 2 seconds.
- Both grippers lower the two ends back onto the desk, still under tension, so the placket lands straight. Release once the shirt is resting.
- Both grippers lift clear.

Nothing is stroked down the placket. A closed gripper dragged along a snapped placket rubs across every snap in turn and peels them open, and on a bare desk it drags the panel beside it out of line as well. Pulling the two ends apart takes the slack out along the closure line instead, which is the direction the snaps are built to hold, so they stay shut.

The pull goes along the placket, never across it. A pull that opens out to the left and right loads the two placket edges away from each other and pops the snaps straight open.

Check: all five snaps are closed, the placket runs straight from the collar to the hem with both edges in line, the front lies flat with no ruck or pucker along the placket, and the shirt is still

square in the work zone with clear desk all round. If a snap sprang open under the tension, close it again as in Step 1 and tension the placket again, more gently. If the front is still rucked, take the two ends again and repeat the pull.

Step 3: Turn the shirt face down by draping it from the shoulders

Goal: the shirt lying face down and square in the work zone, collar toward the front edge, both sleeves flat.

- The **left gripper** pinches the **left shoulder** and the **right gripper** pinches the **right shoulder**.
- Both grippers lift together until the hem hangs clear of the desk and the shirt hangs face up toward the operator.
- Both arms make one **settle shake** so the body and both sleeves drop straight.
- Both grippers carry the hanging shirt toward the operator and down, then move back and down in one smooth motion, so the shirt lays itself over onto its front and the collar lands nearest the front edge. Release once the whole shirt is resting on the desk.
- Both grippers lift clear.
- The **left gripper** pinches the **left cuff**, lifts the left sleeve clear, and lays it out flat angled toward the far left corner. The **right gripper** does the same with the **right cuff** toward the far right corner.

The shirt turns because the arms carry the shoulder hold forward, down, and back past the shirt's own contact line. The wrists do not turn it: forearm rotation is capped near a quarter turn, so a wrist twist cannot bring the shirt over and will stall the arm at its limit.

One settle shake, not a fling. A hard fling loads the closed snaps against their own weight, throws the two long sleeves wherever they land, and can carry the shirt out of the work zone. The shake only lets the cloth hang straight before it goes down.

The sleeves are always laid out after the drape. Long sleeves land folded under the body or doubled back on themselves most times a shirt is turned, and a sleeve left as it landed gets folded into the packet and shows.

Check: the shirt lies face down and square inside the work zone with clear desk all round, collar toward the front edge, hem toward the far side, both sleeves flat on the desk and angled toward the far corners with the cuffs clear of the body, no panel folded under, all five snaps still closed, and no placket edge gaping at the collar or the hem. If it landed crumpled, skewed, or off the work zone, take the two shoulders again and drape it again. If a snap sprang open, the episode is not recoverable by folding over it: close it again as in Step 1, tension the placket again, and drape again.

Step 4: Fold the left side in and lay the left sleeve behind

Goal: the left side folded in on a straight fold edge, with the left sleeve flat inside it.

- The **right gripper**, closed, **pins** the back at the middle of the body and holds it there for the whole step.
- The **left gripper** pinches the **left body edge** at the middle of its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar to the hem, parallel to the placket underneath. Release once it is resting.
- The **left gripper**, closed, strokes along the fold edge from the collar end to the hem end.
- The **left gripper** pinches the **left cuff**, lifts the sleeve clear, carries it back toward the left, and lays it down so the sleeve runs toward the hem end with the cuff pointing at the hem and the whole sleeve inside the fold edge. Release once it is resting.
- The **left gripper**, closed, strokes along the sleeve from the shoulder to the cuff.
- Both grippers lift clear.

The panel is carried over and laid down. A panel dragged across the bare desk rucks the back under it and lands the fold edge off line, and the right gripper's pin is what stops the whole shirt travelling with the stroke.

Check: the fold edge runs straight from the collar to the hem and parallel to the placket, the sleeve lies flat inside the fold edge with the cuff pointing at the hem end, no part of the sleeve or cuff shows past the fold edge or past the hem edge, and there is no ruck along the fold. If the fold edge landed crooked or the panel came over too far, lift that edge, lay it down again, and stroke it.

Step 5: Fold the right side in and lay the right sleeve behind

Goal: the right side folded in on a straight fold edge, with the right sleeve flat inside it, so the body is now one straight sided panel.

- The **left gripper**, closed, **pins** the back at the middle of the body and holds it there for the whole step.
- The **right gripper** pinches the **right body edge** at the middle of its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar to the hem, parallel to the left fold edge. Release once it is resting.
- The **right gripper**, closed, strokes along the fold edge from the collar end to the hem end.
- The **right gripper** pinches the **right cuff**, lifts the sleeve clear, carries it back toward the right, and lays it down so the sleeve runs toward the hem end with the cuff pointing at the hem and the whole sleeve inside the fold edge. Release once it is resting.
- The **right gripper**, closed, strokes along the sleeve from the shoulder to the cuff.
- Both grippers lift clear.

Check: the two fold edges run parallel and the same distance in from the collar on each side, the right sleeve lies flat inside its fold edge with the cuff pointing at the hem end, nothing shows past either fold edge, and the body reads as one straight sided panel the width of the collar. If the two sides came out uneven, lift the right edge, lay it down again to match the left, and stroke it.

Step 6: Rotate the shirt horizontal, collar to the right

Goal: the folded body lying horizontal across the work zone, collar at the right end and hem at the left end, both side fold edges still shut.

- The **right gripper** pinches the **collar end** of the folded body at its middle and the **left gripper** pinches the **hem end** at its middle.
- Both grippers lift the body just clear of the desk, keeping it flat and level so neither side fold opens.
- Both arms swing the body round a quarter turn together, the collar end going to the right and the hem end to the left, and lay it down horizontal in the middle of the work zone. Release once it is resting.
- Both grippers lift clear.

Rotate before the thirds, not after. Horizontal, the shirt's long axis runs left to right in front of the operator, so each arm takes the end nearest it, the thirds folds travel left to right instead of away from the operator, and neither arm has to reach out to the far edge of the desk with a loaded grip.

The two arms carry the body round. The wrists stay where they are: forearm rotation is capped near a quarter turn, so turning the shirt on the wrists alone runs an arm into its limit mid rotation and drops the packet.

Check: the folded body lies horizontal and square in the work zone, collar at the right end, hem at the left end, both side fold edges still shut with no layer sprung open, both sleeves still hidden inside, and clear desk all round. If a side fold sprang open during the rotation, lift that edge, lay it down again, and stroke it. If the body landed at an angle, take the two ends again and lay it down square.

Step 7: Fold the body into thirds

Goal: the body folded into even thirds with the whole collar left showing at the right end.

- The **left gripper** pinches the **far hem corner** and the **right gripper** pinches the **near hem corner**, both at the left end of the body.
- Both grippers lift the hem edge clear, carry it to the right, and lay it down one third of the body's length from the collar, square to the two side fold edges. Release once it is resting.

- The **left gripper**, closed, **pins** the body at its middle. The **right gripper**, closed, strokes along the new left fold edge from its far end to its near end.
- Both grippers pinch that **left fold edge** at its far and near corners, lift it clear, carry it to the right, and lay it down **just short of the collar** so the whole collar stays showing. Release once it is resting.
- The **left gripper**, closed, **pins** the body at its middle. The **right gripper**, closed, strokes along the new left fold edge from its far end to its near end.
- Both grippers lift clear.

Stopping short is the whole point of the second fold. An edge laid over the collar buries it and the shirt does not present.

Check: the shirt is one squared **packet**, both thirds fold edges run parallel to the collar, the thirds are even, the whole collar shows face up at the right end with nothing lying over it, no sleeve or cuff shows past an edge, and no layer has sprung open. If a fold landed crooked or covered the collar, lift that fold edge back up, lay it down again, and stroke it.

Step 8: Confirm the folded shirt and end the episode

- Confirm the packet lies square in the work zone, the whole collar shows face up at the right end, all five snaps are still closed, both sleeves are hidden behind the body with no cuff showing past an edge, no layer has sprung open, and the desk is clear apart from the folded shirt.
- Return both arms home, clear of the work zone, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the folded shirt off the desk and open it out flat.
2. Pull the placket edges apart to open all five snaps. Check every stud and socket is undamaged and still firmly attached, and that each one still clicks closed and holds under a straight press.
3. Close the shirt by hand, lift it by both shoulders, and confirm no snap lets go under its own weight. A shirt whose snaps open when it hangs cannot be draped and is replaced.
4. Press the shirt flat again so no fold line from the last episode is left in the cloth. A shirt that still shows an old fold is not restaged.
5. Replace any shirt that is torn, stained, damp, has an open seam, or carries a snap that will not hold. Vary which shirt is used between episodes rather than running the same one every time.
6. Lay the shirt back face up and square in the work zone, collar toward the far side and hem toward the front edge, both sleeves flat and angled toward the front corners with the cuffs

clear of the placket, and the over placket resting on the under placket with all five snaps open.

7. Wipe the desk dry and clear, and leave it bare apart from the shirt.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Snaps worked out of order

- **Visible cue:** a snap further from the collar is closed while a snap nearer the collar is still open, or the placket is worked from the hem up.
- **SOP rule broken:** Step 1, work the snaps in order, snap 1 at the collar first and snap 5 at the hem last.
- **Coaching note:** start at the collar and count down. The top snap sets the line for the rest.

Violation: Moved on from an open snap

- **Visible cue:** the episode leaves a snap that did not take and goes on to the next snap or to the tension pull, or the placket gapes at a snap position with the two edges apart.
- **SOP rule broken:** Step 1, a snap is closed before the next one is started, and Step 2 begins only with all five closed.
- **Coaching note:** watch it hold when the gripper lifts. If it springs, press it again.

Violation: Snap closed by pinching or sliding

- **Visible cue:** the gripper squeezes the two placket layers between its fingers, presses at an angle, rubs or dabs at the snap, or slides across it instead of setting down flat and holding still.
- **SOP rule broken:** Step 1, one flat press held for 2 seconds and lifted straight off, never pinched and never slid.

- **Coaching note:** down flat, count two, straight up.

Violation: Placket dragged into line

- **Visible cue:** the over placket is pushed, pulled, or shoved sideways to bring a socket onto its stud, instead of being pinched, lifted clear, and lowered onto it.
- **SOP rule broken:** Step 1, lift the over placket clear of the cloth below and lower it onto the stud.
- **Coaching note:** pick it up and put it down. Dragging takes the whole front with it.

Violation: Pressed without a pin

- **Visible cue:** the left gripper is off the cloth while a snap is pressed, it takes the pin only after the press has started, or it lifts away before the right gripper is clear, and the placket shifts under the press.
- **SOP rule broken:** Step 1, the left gripper pins the under placket beside the stud and holds it until the press is finished.
- **Coaching note:** pin first, let go last. Nothing else is holding the placket down.

Violation: Placket stroked instead of tensioned

- **Visible cue:** a closed gripper is pressed onto the placket and drawn down it from the collar toward the hem, or dabbed along it, instead of the two ends being pinched and pulled apart.
- **SOP rule broken:** Step 2, the placket is straightened by pulling the collar end and the hem end apart under tension. Nothing is stroked down the placket.
- **Coaching note:** a gripper dragged down the placket rubs every snap open on the way. Take the two ends and pull.

Violation: Tension skipped, slack, or pulled across the placket

- **Visible cue:** the shirt goes to the turn with no tension pull at all; or both grippers take the two ends but the placket still sags between them and no slack comes out; or the two grippers pull out to the left and right across the placket instead of apart along it, and the placket edges lift away from each other.
- **SOP rule broken:** Step 2, both grippers come to the same height and draw apart along the collar to hem line until the placket hangs straight, never across it.
- **Coaching note:** pull the way the placket runs, top away from bottom. Pull it sideways and the snaps come open.

Violation: Folded while still face up

- **Visible cue:** a side, a sleeve, or the hem is folded before the shirt has been turned, so a fold is made with the placket still showing.
- **SOP rule broken:** Step 3, the shirt is turned face down before anything is folded.

- **Coaching note:** fasten, tension, turn. Nothing folds until the front is against the desk.

Violation: Turned by the wrong hold or dragged over

- **Visible cue:** the shirt is turned by its hem corners, by one shoulder, by a sleeve, or by the collar; or it is rolled, shoved, or walked over on the desk without being lifted clear; or a wrist twists to bring it over instead of the arms carrying the hold forward and back.
- **SOP rule broken:** Step 3, both grippers take the two shoulders, lift until the shirt hangs clear, and carry the hold forward, down, and back to drape it over.
- **Coaching note:** both shoulders, lift it off the desk, and walk it over. The wrists do not have the range to turn it.

Violation: Thrown instead of draped

- **Visible cue:** the shirt is whipped, flung, swung, or shaken repeatedly while it hangs; it is released before it is resting and drops the last part of the way; or it lands crumpled, folded on itself, or outside the work zone.
- **SOP rule broken:** Step 3, one settle shake, then one smooth carry forward, down, and back, and release only once the whole shirt is resting.
- **Coaching note:** one shake to let it hang straight, then lay it over. A fling pops snaps and throws the sleeves.

Violation: Sleeves left as they landed after the turn

- **Visible cue:** the episode goes on to the side folds with a sleeve still folded under the body, doubled back on itself, lying across the back, or hanging off the work zone, without each cuff being lifted and laid out flat.
- **SOP rule broken:** Step 3, both cuffs are lifted and both sleeves laid out flat and angled toward the far corners before anything is folded.
- **Coaching note:** long sleeves never land right. Lay both of them out every time.

Violation: Fold made by dragging

- **Visible cue:** a body edge, a cuff, or the hem edge is pushed, shoved, or slid across the desk into place instead of being pinched, lifted clear, carried over, and lowered onto its target.
- **SOP rule broken:** Steps 4, 5, and 7, lift the edge clear before carrying it.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Side fold edge off line

- **Visible cue:** a side fold edge runs crooked instead of straight from the collar to the hem, is not parallel to the other side, or the panel comes over so far that it crosses the placket line or so little that the body is left wider than the collar.

- **SOP rule broken:** Steps 4 and 5, the fold edge runs straight from the outer edge of the collar to the hem, parallel to the placket and to the other fold edge.
- **Coaching note:** line the edge up on the collar before you lay it down.

Violation: Sleeve left showing

- **Visible cue:** a cuff or part of a sleeve sticks out past a side fold edge or past the hem edge, a sleeve is left lying across the front, or a sleeve is folded but never laid flat and stays bunched under the fold.
- **SOP rule broken:** Steps 4 and 5, the whole sleeve lies flat inside the fold edge with the cuff pointing at the hem end.
- **Coaching note:** the sleeve lives behind the body. Nothing of it shows from the front.

Violation: Stroke skipped or dabbed

- **Visible cue:** a side fold edge, a sleeve, or a thirds fold edge is left with no stroke drawn along it, or the gripper taps it in spots or presses once in the middle instead of drawing one continuous run end to end.
- **SOP rule broken:** Steps 4, 5, and 7, one continuous stroke end to end without lifting off.
- **Coaching note:** one run, end to end, without lifting off.

Violation: Stroked without a pin

- **Visible cue:** a fold edge or a thirds fold edge is stroked with the other gripper off the cloth, and the shirt travels, slides, or skews across the bare desk under the stroke.
- **SOP rule broken:** Steps 4, 5, and 7, one gripper pins the body while the other strokes.
- **Coaching note:** hold it down before you smooth it. Bare desk lets the whole shirt go with you.

Violation: Thirds folded without rotating horizontal

- **Visible cue:** the hem end is folded toward the collar while the shirt still lies upright with the collar toward the front edge, so the thirds folds travel away from the operator instead of left to right.
- **SOP rule broken:** Step 6, the shirt is rotated horizontal with the collar at the right end before the thirds are folded.
- **Coaching note:** turn it sideways first. Collar right, hem left, then fold left to right.

Violation: Rotation dragged or sprang a fold

- **Visible cue:** the folded body is pushed, swept, or spun round on the desk instead of being lifted clear by its two ends; it is rotated by one gripper alone or by a wrist twist; or it lands with a side fold edge sprung open, a sleeve out, or the body at an angle.

- **SOP rule broken:** Step 6, both grippers take the collar end and the hem end, lift the body clear and level, swing it round together, and lay it down horizontal and square.
- **Coaching note:** both ends, up, round, down. Keep it flat so nothing opens on the way.

Violation: Thirds fold buried the collar

- **Visible cue:** the second thirds fold is laid over the collar or onto the collar band, so the collar is covered or only part of it shows in the finished packet.
- **SOP rule broken:** Step 7, the second fold stops just short of the collar so the whole collar stays showing.
- **Coaching note:** stop short. The collar is what the fold presents.

Violation: Thirds uneven or out of square

- **Visible cue:** the hem edge lands well past or well short of the third line, a fold edge runs at an angle to the collar instead of parallel to it, or the finished packet's sides step out of line with each other.
- **SOP rule broken:** Step 7, the body is folded into even thirds, square to the side fold edges.
- **Coaching note:** take both corners together and keep the edge square as it comes over.

Violation: Regripped without laying the shirt down

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on the shirt, a panel, a shoulder, or a cuff without laying it down first, including a shoulder hold reseated while the shirt is hanging.
- **SOP rule broken:** Steps 1 to 7, take a fresh hold only after the cloth is resting.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens above the desk and the cloth falls the rest of the way, or it opens over bare desk outside the work zone or over open space.
- **SOP rule broken:** Steps 1 to 7, the gripper opens in the work zone once the cloth is resting, and nowhere else.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, brushes, or straightens a panel, a sleeve, or the packet after releasing it, instead of lifting that edge clear and laying it down again.
- **SOP rule broken:** Steps 3 to 7, correct a bad fold by lifting that edge and laying it down again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Shirt dropped, collided, or knocked off the work zone

- **Visible cue:** the shirt or a sleeve falls out of the work zone or off the desk, the shirt is dropped from the shoulder hold, a finished fold comes apart in transit, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, keep the shirt inside the work zone on a clear path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a snap open, a sleeve showing, a layer sprung open, the collar covered or face down, the collar at the wrong end, the shirt still spread on the desk, or an arm away from home.
- **SOP rule broken:** Step 8, confirm the squared packet, the five closed snaps, the hidden sleeves, and the whole collar showing face up at the right end; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective snap:** will not click closed under three correct presses, pops open on its own while the shirt is untouched, or lets go under the shirt's own weight when it hangs from the shoulders during a correct drape. Replace the shirt before the next episode.
- **Shirt will not hold a fold:** stroked end to end and the fold still springs open every time. Press the shirt before the next episode.
- **Defective shirt:** found torn, stained, damp, with an open seam, or with a placket that will not lie flat.
- **Shirt is too large for the cell,** so it cannot lie spread inside the work zone with clear desk all round, a cuff sits outside an arm's reach when the shirt is staged, or the hem still touches the desk when the shirt hangs from both shoulders.
- **Shirt too slippery for the bare desk:** it skates out of position under a correct pin every time, so no fold edge can be held.

Annotation subtasks (from SOP)

1. Press one snap closed on the placket
2. Pull the placket straight under tension between both grippers

3. Drape the shirt face down from both shoulders
4. Lay one sleeve out flat after the turn
5. Fold one side of the body in
6. Lay one sleeve flat behind the body
7. Rotate the shirt horizontal with the collar to the right
8. Fold the hem end across one third
9. Return both arms home and end the episode